through the use of a periodic impulse train multiplied by the continuous-time signal $x(t)$ that we wish to sample. This mechanism, known as *impulse-train sampling,* is depicted in Figure 7.2. The periodic impulse train $p(t)$ is referred to as the *sampling function,* the period T as the *sampling period,* and the fundamental frequency of $p(t)$, $\omega_s = 2\pi/T$, as the *sampling frequency.* In the time domain,

$$x_p(t) = x(t)p(t), \tag{7.1}$$

where

$$p(t) = \sum_{n=-\infty}^{+\infty} \delta(t - nT). \tag{7.2}$$

Because of the sampling property of the unit impulse discussed in Section 1.4.2, we know that multiplying $x(t)$ by a unit impulse samples the value of the signal at the point at which the impulse is located; i.e., $x(t)\delta(t - t_0) = x(t_0)\delta(t - t_0)$. Applying this to eq. (7.1), we see, as illustrated in Figure 7.2, that $x_p(t)$ is an impulse train with the amplitudes o

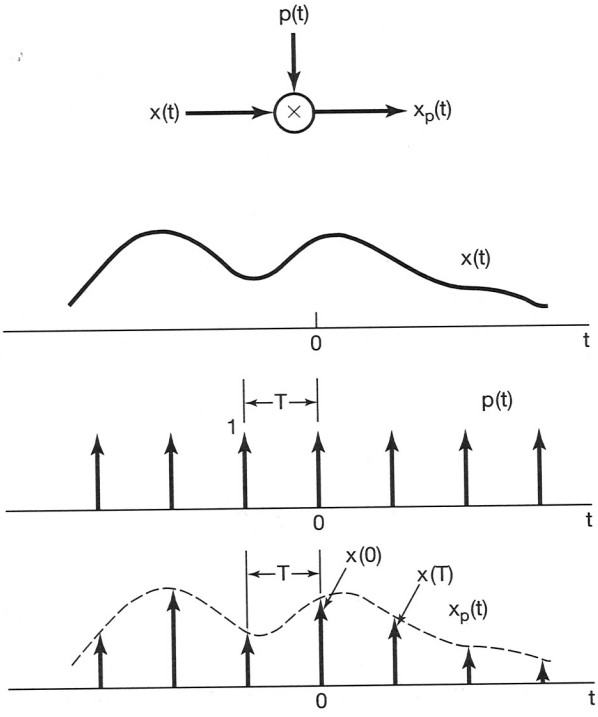


Figure 7.2 Impulse-train sampli

nvenient way in which to represent
ervals. A useful way to do this is
by the continuous-time signal $x(t)$
pulse-train sampling, is depicted
d to as the *sampling function*, the
frequency of $p(t)$, $\omega_s = 2\pi/T$, as

(7.1)

(7.2)

e discussed in Section 1.4.2, we
value of the signal at the point at
$-t_0$). Applying this to eq. (7.1),
lse train with the amplitudes of

7.2 Impulse-train sampling.

the impulses equal to the samples of $x(t)$ at intervals spaced by T; that is,

$$x_p(t) = \sum_{n=-\infty}^{+\infty} x(nT)\delta(t - nT). \tag{7.3}$$

From the multiplication property (Section 4.5), we know that

$$X_p(j\omega) = \frac{1}{2\pi}\int_{-\infty}^{+\infty} X(j\theta)P(j(\omega - \theta))d\theta. \tag{7.4}$$

and from Example 4.8,

$$P(j\omega) = \frac{2\pi}{T}\sum_{k=-\infty}^{+\infty} \delta(\omega - k\omega_s). \tag{7.5}$$

Since convolution with an impulse simply shifts a signal [i.e., $X(j\omega) * \delta(\omega - \omega_0) = X(j(\omega - \omega_0))$], it follows that

$$X_p(j\omega) = \frac{1}{T}\sum_{k=-\infty}^{+\infty} X(j(\omega - k\omega_s)). \tag{7.6}$$

That is, $X_p(j\omega)$ is a periodic function of ω consisting of a superposition of shifted replicas of $X(j\omega)$, scaled by $1/T$, as illustrated in Figure 7.3. In Figure 7.3(c), $\omega_M < (\omega_s - \omega_M)$, or equivalently, $\omega_s > 2\omega_M$, and thus there is no overlap between the shifted replicas of $X(j\omega)$, whereas in Figure 7.3(d), with $\omega_s < 2\omega_M$, there is overlap. For the case illustrated in Figure 7.3(c), $X(j\omega)$ is faithfully reproduced at integer multiples of the sampling frequency. Consequently, if $\omega_s > 2\omega_M$, $x(t)$ can be recovered exactly from $x_p(t)$ by means of

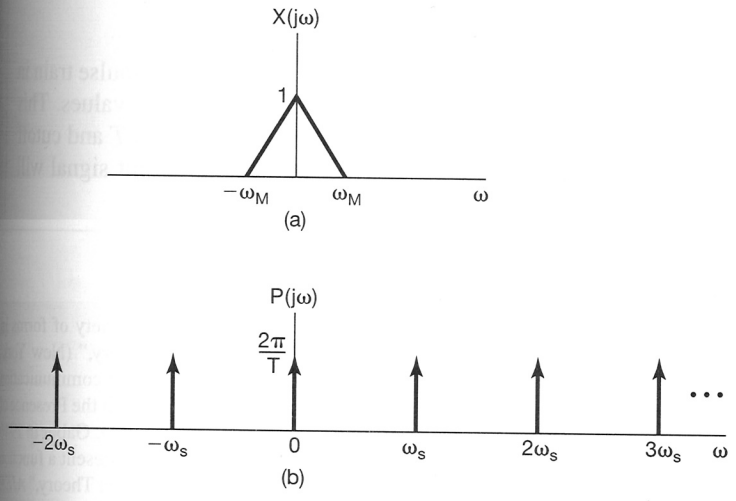


Figure 7.3 Effect in the frequency domain of sampling in the time domain: (a) spectrum of original signal; (b) spectrum of sampling function;

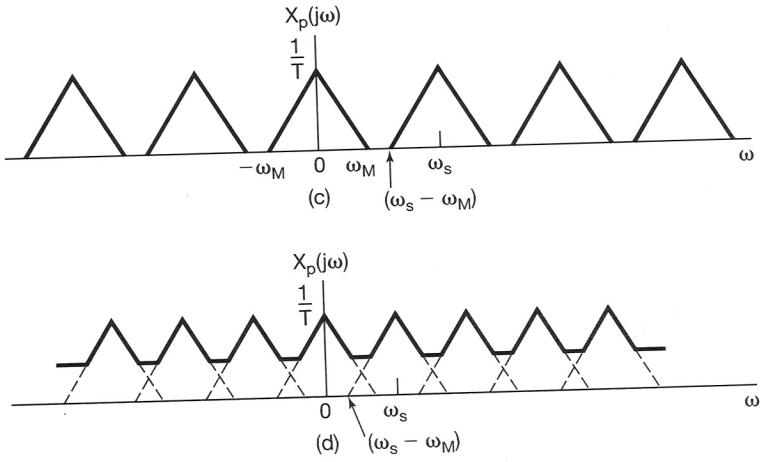


Figure 7.3 *Contin*
of sampled signal wit
(d) spectrum of sam
$\omega_s < 2\omega_M$.

a lowpass filter with gain T and a cutoff frequency greater than ω_M and le
as indicated in Figure 7.4. This basic result, referred to as the *sampling*
stated as follows: [1]

Sampling Theorem:
Let $x(t)$ be a band-limited signal with $X(j\omega) = 0$ for $|\omega| > \omega_M$. Then
determined by its samples $x(nT)$, $n = 0, \pm 1, \pm 2, \ldots$, if

$$\omega_s > 2\omega_M,$$

where

$$\omega_s = \frac{2\pi}{T}.$$

Given these samples, we can reconstruct $x(t)$ by generating a periodic
which successive impulses have amplitudes that are successive sam
impulse train is then processed through an ideal lowpass filter with g
frequency greater than ω_M and less than $\omega_s - \omega_M$. The resulting ou
exactly equal $x(t)$.

[1]The important and elegant sampling theorem was available for many years i
the mathematics literature. See, for example, J. M. Whittaker, "Interpolatory Functio
Stecher-Hafner Service Agency, 1964), chap. 4. It did not appear explicitly in the litera
theory until the publication in 1949 of the classic paper by Shannon entitled "Communic
Noise" (*Proceedings of the IRE,* January 1949, pp. 10–21). However, H. Nyquist in 192
had pointed out, based on the use of the Fourier Series, that $2TW$ numbers are sufficient
of duration T and highest frequency W. [H. Nyquist, "Certain Topics in Telegraph Trans
Transactions, 1928, p. 617; D. Gabor, "Theory of Communication," *Journal of IEE* 93,